

5.2.2 Work Integrals_Guide

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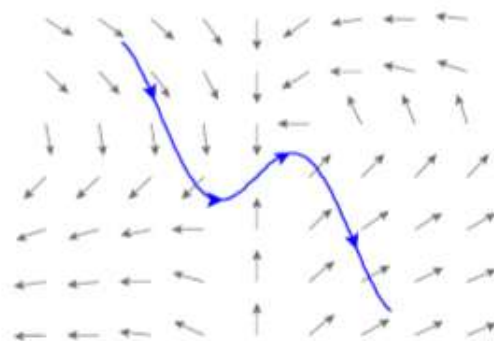


5.2.2 Work
Integrals_...

5.2.2 Work Integrals

Motivation: Suppose, as in the last section, we have a curve C that we are interested in travelling along, but we are now wanting to move along this curve through a vector field.

In the last section we saw that a line integral was useful for “adding up” a quantity as we progressed along a path. In this section we are interested in the vector field’s effect on the motion of an object along the curve C , specifically in regards to the amount of work done by the vector field on any object that traverses C .



When the vector field is moving with the direction of the curve it contributes positive work in the motion of an object along the curve. When the vector field is moving against the direction of the curve it contributes negative work in the motion of an object along the curve.

Exploration: Suppose that $\mathbf{F}(x, y, z) = \langle P(x, y, z), Q(x, y, z), R(x, y, z) \rangle$ is a continuous vector field.

Suppose also that $\mathbf{F}(x, y, z)$ models some force field in space (for example $\mathbf{F}$ could represent a gravitational field or an electric force field).

Suppose a particle moves through this force field along a smooth curve C which is given by the parameterization $\mathbf{r}(t)$ where $a \leq t \leq b$. Suppose that the particle starts its movement at the point P_0 and ends its movement at the point P_n . We want to know the amount of work done by $\mathbf{F}$ on the particle as it moves from P_0 to P_n .

We start by partitioning $[a, b]$ into n equally spaced subintervals

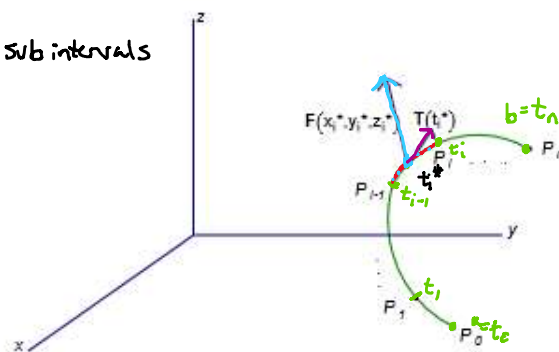
Let Δs_i represent the length of P_{i-1} to P_i

let t_i^* be some sample point in $[t_{i-1}, t_i]$

and suppose $\vec{r}(t_i^*) = \langle x_i^*, y_i^*, z_i^* \rangle$

We approximate the work done by $\vec{F}$ on the particle over $[t_{i-1}, t_i]$

by using $\vec{F}(x_i^*, y_i^*, z_i^*)$



We can also approximate the direction the particle is moving over $[t_{i-1}, t_i]$ by using $\vec{T}(t_i^*)$

So we can estimate then that the "amount" of $\vec{F}$ that "pushes" the particle along C is given by

$$\vec{F}(x_i^*, y_i^*, z_i^*) \cdot \vec{T}(t_i^*)$$

So, by the way work is defined, the work done by $\vec{F}$ on the particle as the particle moves from P_{i-1} to P_i is approximately:

$$W_i = \vec{F} \cdot d = \vec{F}(x_i^*, y_i^*, z_i^*) \cdot \vec{T}(t_i^*) \cdot \Delta s_i$$

Force
distance
dot product
multiplication

Thus, the total work done by the force $\vec{F}$ in moving the particle along C is:

$$W = \lim_{n \rightarrow \infty} \sum_{i=1}^n \vec{F}(x_i^*, y_i^*, z_i^*) \cdot \vec{T}(t_i^*) \cdot \Delta s_i = \int_C \vec{F} \cdot \vec{T} ds$$

We know that $ds = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt = \|\vec{r}'(t)\| dt$, and from chapter 2 we know that $\vec{T}(t) = \frac{\vec{r}'(t)}{\|\vec{r}'(t)\|}$.

So, we have:

$$\int_C \vec{F} \cdot \vec{T} ds = \int_C \vec{F}(\vec{r}(t)) \cdot \frac{\vec{r}'(t)}{\|\vec{r}'(t)\|} \cdot \|\vec{r}'(t)\| dt = \int_C \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) dt$$

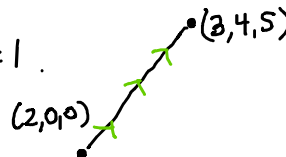
Definition: Let $\vec{F}$ be a continuous vector field defined on a smooth curve C given by a vector function $\vec{r}(t)$ where $a \leq t \leq b$. Then, the line integral of $\vec{F}$ along C is

$$\int_C (\vec{F} \cdot \vec{T}) ds = \int_a^b (\vec{F}(\vec{r}(t)) \cdot \vec{r}'(t)) dt = \int_C \vec{F} \cdot d\vec{r}$$

Note: We will address an intuitive meaning of $d\vec{r}$ in the next example. One should note that $\vec{F}$ may have any number of components and the above definition will still apply.

Example 1: Find the work done by the force field $\mathbf{F} = \langle y, z, x \rangle$ in moving a particle along the line segment, C , from $(2, 0, 0)$ to $(3, 4, 5)$. (Notice this is the same as Ex 5 in section 5.2.1!)

We parametrize C as $\vec{r}(t) = \langle \underline{t+2}, \underline{4t}, \underline{5t} \rangle$ for $0 \leq t \leq 1$.



$$W = \int_C (\vec{F} \cdot \vec{T}) ds = \int_C \vec{F} \cdot d\vec{r} = \int_0^1 \langle 4t, 5t, t+2 \rangle \cdot \langle 1, 4, 5 \rangle dt$$

$$= \int_0^1 (4t + 20t + (t+2)5) dt$$

$$= \int_0^1 (29t + 10) dt$$

$$= \left. \frac{29}{2}t^2 + 10t \right|_0^1$$

$$= \left(\frac{29}{2} + 10 \right) - 0 = \frac{49}{2}$$

Observation:

We can view our resulting work integral as $W = \int_C y dx + z dy + x dz$.

We can think of $d\vec{r} = \langle dx, dy, dz \rangle$

$$\text{Then } W = \int_C y dx + z dy + x dz = \int_C \langle y, z, x \rangle \cdot \langle dx, dy, dz \rangle$$

$$= \int_C \vec{F} \cdot d\vec{r}$$

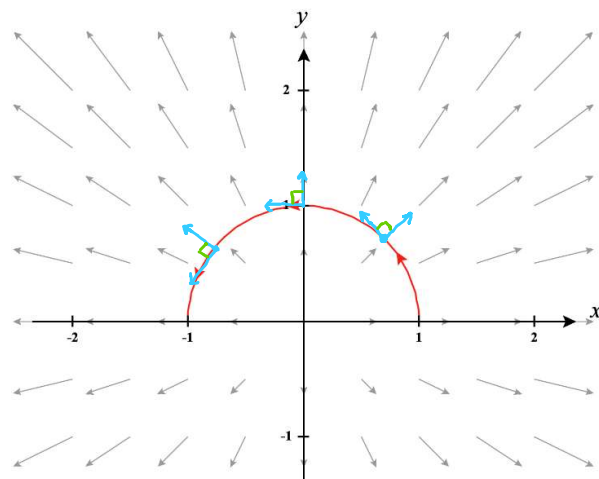
Example 2: Find the work done by $\mathbf{F}(x, y) = \langle x, y \rangle$ on a particle which moves along the curve C given by

$$\mathbf{r}(t) = \left\langle \frac{\cos(t)}{x}, \frac{\sin(t)}{y} \right\rangle \text{ where } 0 \leq t \leq \pi. \quad \begin{array}{l} dx = -\sin(t) dt \\ dy = \cos(t) dt \end{array}$$

$$\begin{aligned} W &= \int_C \vec{F} \cdot d\vec{r} = \int_C \langle x, y \rangle \cdot \langle dx, dy \rangle \\ &= \int_C x dx + y dy = \int_C x dx + \int_C y dy \\ &= \int_0^\pi (-\cos(t)\sin(t) dt + \sin(t)\cos(t) dt) \\ &= \int_0^\pi (-\cancel{\cos(t)\sin(t)} + \sin(t)\cos(t)) dt \\ &= 0 \end{aligned}$$

OR

$$\begin{aligned} W &= \int_C \vec{F} \cdot d\vec{r} = \int_0^\pi \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) dt = \int_0^\pi \langle \cos(t), \sin(t) \rangle \cdot \langle -\sin(t), \cos(t) \rangle dt \\ &= \int_0^\pi (-\cancel{\cos(t)\sin(t)} + \sin(t)\cos(t)) dt \\ &= 0 \end{aligned}$$



Recall

$$\int_C \vec{F} d\vec{r} = \int_C (\vec{F} \cdot \underline{\vec{T}}) ds = \int_C (0) ds = 0$$

Note: One should draw a picture of the particle's path and this vector field to see why the work is equal to 0.

Note: One could have solved the above problem with $W = \int_0^\pi \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt$ as well.